Course Syllabus

Department of Mechanical Engineering
Chulalongkorn University

Academic Year: Fall 2026 (First Semester)

Course Information

Course Number	2103304
Course Title	Automatic Control I
Credits	3 (3–0–9)
Semester	Fall 2026 (First Semester)
Instructor	Chawin Ophaswongse, Ph.D.
Co-Instructor	Assoc. Prof. Thitima Jintanawan, Ph.D.
Teaching Assistants (TAs)	...
Class Time	Monday, 13:00–16:00
Location	Room 409, Building 3, Faculty of Engineering
Office Hours	TBD, Hans Bunthli Building, 4th Floor, Room 407
Prerequisite	...

Course Description

This course provides a foundational understanding of feedback control systems with applications in mechanical engineering. It develops the full analysis-and-design workflow: representing a physical system, deriving its governing equations, and building the engineering models needed to predict and shape its behavior. Core representations include ordinary differential equations, transfer functions, and state-space models, developed from mechanical and electro-mechanical examples such as mass–spring–damper systems and DC-motor drives. Models are obtained both from first principles and experimentally through *system identification*, fitting a model to measured input–output data.

Students learn to analyze system behavior in both the time and frequency domains (transient response, steady-state error, and stability) and to design and tune controllers (P, PI, PD, PID, lead/lag, and full-state feedback) to meet performance specifications such as rise time, overshoot, and settling time. Classical techniques including block-diagram reduction, the Routh–Hurwitz criterion, root locus, and Bode/Nyquist frequency response are introduced alongside an entry-level treatment of state-space design.

Analysis and design are practiced throughout using **MATLAB/Simulink** with the Control System Toolbox. The *Servo Mini-labs* and a *Final Project* connect theory to hardware through a MATLAB script or Simulink-based control interface to a real servo plant (a brushed DC-motor system). The emphasis is on control *analysis and design*: identifying the plant model from experiment (system identification), designing and tuning the controller, and interpreting the measured response, rather than low-level firmware or embedded programming, so students focus on control-engineering decisions instead of implementation plumbing.

Course Learning Outcomes (CLOs)

On successful completion of this course, students will be able to:

CLO	Course Learning Outcome	PI
1	Identify the relevant parameters and derive the governing equations of a mechanical or electro-mechanical system, and lay out a systematic procedure for analyzing a real engineering problem (equations of motion, transfer functions, free/damped vibration, stability).	1.2
2	Construct engineering models of a given problem or real system suitable for analysis (free-body diagrams, kinematics/kinetics diagrams, control block diagrams) and translate them into differential-equation, transfer-function, and state-space representations.	1.3
3	Analyze system behavior in the time and frequency domains, including transient response, steady-state error, and stability.	1.2
4	Design and tune feedback controllers (P/PI/PD/PID, lead/lag, full-state feedback) to meet performance specifications, using root locus and frequency-response methods.	1.2
5	Apply MATLAB/Simulink to identify, model, simulate, analyze, and design control systems, and validate a controller on a real servo plant through a Simulink hardware interface.	1.3

Program Performance Indicators (PIs) Addressed

- **PI 1.2 (Parameters and governing equations in applied mechanics).** Ability to identify variables, write the governing (control) equations, and define the procedure for analyzing engineering problems and real mechanical systems (e.g., static equilibrium, stress–strain in beams, stress transformation, failure theory, equation of motion, transfer function, free/damped vibration, and stability).
- **PI 1.3 (Engineering modeling).** Ability to build mechanical-engineering models of problems and real systems for use in analysis (e.g., free-body diagram, kinematics diagram, kinetics diagram, stress–strain element, control block diagram, P–V diagram, flow diagram, and energy-transfer diagram).

Textbooks

Primary

- Nise, N. S. (2011). *Control Systems Engineering* (6th ed.). Wiley.
  (Beginner-friendly)

Supplementary

- Ogata, K. (2010). *Modern Control Engineering* (5th ed.). Prentice Hall.
  (State space; MATLAB; more mathematically rigorous)
- Dorf, R. C., & Bishop, R. H. (2017). *Modern Control Systems* (13th ed.). Pearson.
  (Design emphasis; rich in illustrative examples)

Course Schedule

Wk	Date	Topic	Key Content	HW	Servo Mini-lab
1	3 Aug	Intro & Math Primer	Open-/closed-loop · ODEs · Laplace	HW1	
2	10 Aug	System Modeling I	LTI · transfer functions · elec./mech.		Lab 1: Intro & Setup
3	17 Aug	System Modeling II	State space · nonlinearities · linearization	HW2	
4	24 Aug	Time Response	1st/2nd-order · transient-response specs	HW3	Lab 2: Time Response
5	31 Aug	Block Diagrams	Reduction · equivalent forms · feedback		
6	7 Sep	Stability	Routh–Hurwitz · gain range for stability	HW4	Lab 3: P-Control & Stability
7	14 Sep	Midterm Review	Review of Weeks 1–6		
Midterm Examination: Wednesday, 23 September 2026, 13:00–16:00					
8	28 Sep	Steady-State Errors	System type · gain design · disturbances	HW5	
9	5 Oct	Root Locus I	Sketching rules · transient response via gain	HW6	
10	12 Oct	Root Locus II	PI/PD/PID · lead/lag compensation		Lab 4: Root Locus
11	19 Oct	Frequency Response I	Asymptotic Bode plots · bandwidth	HW7	
12	26 Oct	Frequency Response II	Nyquist & margins · lead/lag · PID		Lab 5: Frequency Response
13	2 Nov	Design via State Space	Controllability · full-state feedback	HW8	Lab 6: Pole Placement
14	9 Nov	Project Demo Day	Group presentations · control competition		
15	16 Nov	Final Review	Review of all topics		
Final Examination: Monday, 30 November 2026, 08:30–11:30					

Course Policies

- **Communication.** Announcements and materials are posted on MyCourseVille; day-to-day discussion is on the course Discord.
- **Attendance.** Mini-lab sessions are mandatory and graded.
- **Mini-labs.** Pre-lab handouts are released one week before each mini-lab (reading required before the session). Lab reports (groups of up to 4) are due one week after the lab session.
- **Assignments.** Homework is due 10 days after it is assigned unless otherwise specified. Late work is penalized per the stated policy.
- **Academic integrity.** Collaboration is encouraged, but submitted work must be each group's own; plagiarism and exam misconduct follow Faculty of Engineering regulations.
- **Software.** Install MATLAB/Simulink and the Control System Toolbox before Week 2.

Assessment

Component	Weight
Assignments (Homework)	10%
Servo Mini-labs	10%
Midterm Examination	30%
Final Project	15%
Final Examination	35%
Total	**100%**

Servo Mini-labs

Six 45-minute hands-on sessions run in parallel with the lectures. Students identify a plant model from experiment (system identification), model its components (actuator, sensor, controller), then design and tune controllers in **MATLAB/Simulink** and validate them on a real-time servo setup: a brushed DC-motor with an interchangeable inertia-disk or pendulum load. Interaction with the hardware is entirely through the Simulink control interface (*no low-level or embedded programming is required*), so the focus stays on control analysis and design. Sessions are mandatory and graded; a laptop with MATLAB/Simulink and the Control System Toolbox is required.

No-load vs. inertia-disk comparison (Labs 2–6). The inertia disk is used from Lab 2 onward as a single-parameter change to the plant: mounting it raises the inertia J, which scales the mechanical time constant $\tau = J/(b + K_t K_b/R)$ while leaving the DC gain unchanged. Each session therefore begins with a short no-load baseline test on the bare shaft, repeated with the disk attached, and every lab report includes an overlay of the two responses with the resulting parameter change quantified.

Tentative session plan (subject to adjustment):

- **Lab 1: Intro & Setup (Wk 2).** Connect the DC-motor/inertia-disk plant to Simulink; get familiar with the control interface, encoder sensing, and actuation; run open-loop step tests and a first system-identification of the motor parameters (no-load baseline).
- **Lab 2: Time Response (Wk 4).** Measure open- and closed-loop step responses; extract first-/second-order parameters and transient specifications (rise time, overshoot, settling time) and compare against the identified model. *No-load vs. disk:* the steady-state speed is unchanged while τ increases; cross-check the identified inertia against the geometric value $J_d = \frac{1}{2}mr^2$ and the predicted $\tau_{\text{disk}} = \tau_0(1 + J_d/J_0)$.
- **Lab 3: P-Control & Stability (Wk 6).** Implement proportional position/speed control; observe how gain shapes transient response and the onset of instability. *No-load vs. disk:* hold K_p fixed across both conditions and explain the increased overshoot and ringing from $\omega_n \propto 1/\sqrt{J}$ and $\zeta \propto 1/\sqrt{J}$.
- **Lab 4: Root Locus (Wk 10).** Design controller gain (P/PD/PID) via root locus to meet transient specifications and validate on the plant. *No-load vs. disk:* apply the gain designed for the loaded plant to the unloaded plant and observe the specification violation (motivating robustness).
- **Lab 5: Frequency Response (Wk 12).** Measure the plant's frequency response (Bode); read gain/phase margins and tune a lead/lag or PID controller. *No-load vs. disk:* compare bandwidth and the gain/phase margins of a fixed controller under both load conditions.

- **Lab 6: Pole Placement (Wk 13).** Design full-state feedback by pole placement for the DC-motor with the pendulum load and validate the closed-loop response. *No-load vs. disk:* design for each load condition separately and show that feedback can render the two closed-loop responses nearly identical, subject to the available actuator authority.

Final Project

Working in groups of up to four (beginning after the midterm), students carry out the complete workflow on a given electro-mechanical plant: performing system identification to obtain the plant model, building a **Simulink simulation model** and validating it against the measured hardware response, then designing, tuning, and demonstrating a controller to meet target performance specifications through the MATLAB/Simulink hardware interface. Comparing simulation against hardware is a central theme: students are expected to explain and reconcile the differences they observe.

Tentative milestones (subject to adjustment):

- **Milestone 1: System ID & Simulation Model (Week 9, 5 Oct).** Run experiments to identify the plant parameters; build a Simulink simulation model and validate it against the measured open-loop hardware response.
- **Milestone 2: Controller Design in Simulation (Week 11, 19 Oct).** Specify target performance, design and tune the controller in simulation, and verify that it meets the specifications on the simulation model.
- **Milestone 3: Hardware Implementation & Tuning (Week 13, 2 Nov).** Deploy the controller on the real plant, retune as needed, and document the discrepancies between simulation and hardware.
- **Final Deliverable: Demo Day & Report (Week 14, 9 Nov).** Present the final closed-loop performance in a demo-day competition and submit the final project report.

This syllabus is subject to minor adjustment; any changes will be announced in class and on MyCourseVille.